

Cyril Kafle

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EDUCATION

Northern Virginia Community College

Woodbridge, VA

Associate of Science (Electrical Engineering) | GPA: 3.9

Aug. 2025 – Dec. 2026

- **Coursework:** Calculus I & II, Calculus III, Computer Programming for Engineers, Electric Circuits I & II, Differential Equations, University Physics I & II, Computer Engineering, Engineering Design

- University of Virginia: Guaranteed Admission Agreement (pending Fall 2026 completion); transferring Fall 2027

Thomas Jefferson High School for Science and Technology

Alexandria, VA

Accepted Into #1 Nationally-Ranked Public HS | GPA: 3.6

Aug. 2021 – Jun. 2025

- EVGP Team Member | Prototyping and Engineering Team Captain | Senior Research 2025 Lead Captain

EXPERIENCE

Electronics Repair Associate

Jun. 2022 – Jul. 2022

Self Employed Family Business, Electronics and Trading

Kathmandu, Nepal

- Diagnosed and repaired consumer electronics and control boards for a family-owned electronics business.

Line Operator and Cashier

Apr. 2025 – Jun. 2025

Chipotle Mexican Grill

Lorton, VA

- Delivered line and register service during peak hours, using trilingual communication skills to support customers and coworkers.

PROJECTS

PCBInsight AI | Automated PCB Design Review Platform

Jul. 2026

- Built a full-stack PCB design review platform with a custom KiCad file parser and a deterministic 28-check engineering analysis engine spanning routing, power, thermal, and manufacturability.
- Designed an evidence-constrained AI review layer where citations are code-enforced against deterministic findings, validated with a 10-board test corpus and live testing against production Claude models.
- Developed an interactive board visualizer, automated PDF reporting, and a FastAPI + React dashboard, backed by 147 automated tests and CI/CD.
- Shipped v1.0.0 with a public landing page and live demo; validated the AI layer against production Claude models with zero hallucinated citations across a 10-board engineering validation corpus.

Built with Python (FastAPI), React/TypeScript, Claude API

Polaris | Automated B2B Sales Pipeline

May 2026

- Built the contact and social discovery service: scraped prospect websites for email and social links, implemented Facebook Graph API verification for business pages, and added a web-search fallback; queued leads for SMS outreach when no digital contact was found.
- Rewrote the mockup template engine (~1,500 lines) that auto-generates branded, niche-specific landing pages populated with real business data (name, Google reviews, photos) from AI-generated lead briefs, replacing a prospect's cold pitch with a finished site demo.
- Implemented business-hours data migration and enrichment logic, and hardened the AI lead-scoring pipeline across an end-to-end system spanning lead discovery, enrichment, mockup generation, outreach, and reply monitoring.
- Collaborated with a co-founder in a 2-person team to ship the platform end-to-end, from lead generation through automated outreach and reply monitoring.

Built with Next.js 15, React 19, Supabase (Postgres)

Multifunctional Folding Furniture | TJHSST Senior Research

Aug. 2024 – Jun. 2025

- Designed and built a multifunctional folding furniture system using the full engineering design process: from initial sketches to CAD modeling, CAM toolpath creation, and CNC fabrication in Fusion 360.
- Measured, sourced, and machined raw wood and other materials; performed CNC routing, welding, bandsaw cutting, chiseling, and joint construction to assemble a fully functional prototype.
- Developed a space-efficient, low-cost furniture solution aimed at improving accessibility for people with limited living space or financial constraints.

TECHNICAL SKILLS

Languages: English, Nepali, Spanish

Programming Languages: Python, C++, Java, JavaScript, TypeScript, SQL

Software & Engineering Tools: Autodesk Fusion 360, KiCad, Next.js / React, Pydantic, ReportLab, Matplotlib, Playwright, Zod, Tailwind CSS, FastAPI, Autodesk Inventor, Onshape, MecAgent, Google Places API, Gmail API, Facebook Graph API, PrusaSlicer / Prusa3D, Anthropic Claude API, Supabase (PostgreSQL)

Developer Tools: Git / GitHub, GitHub Actions (CI/CD), Vite, pytest, Arduino IDE, Raspberry Pi, RStudio, TI MSP430